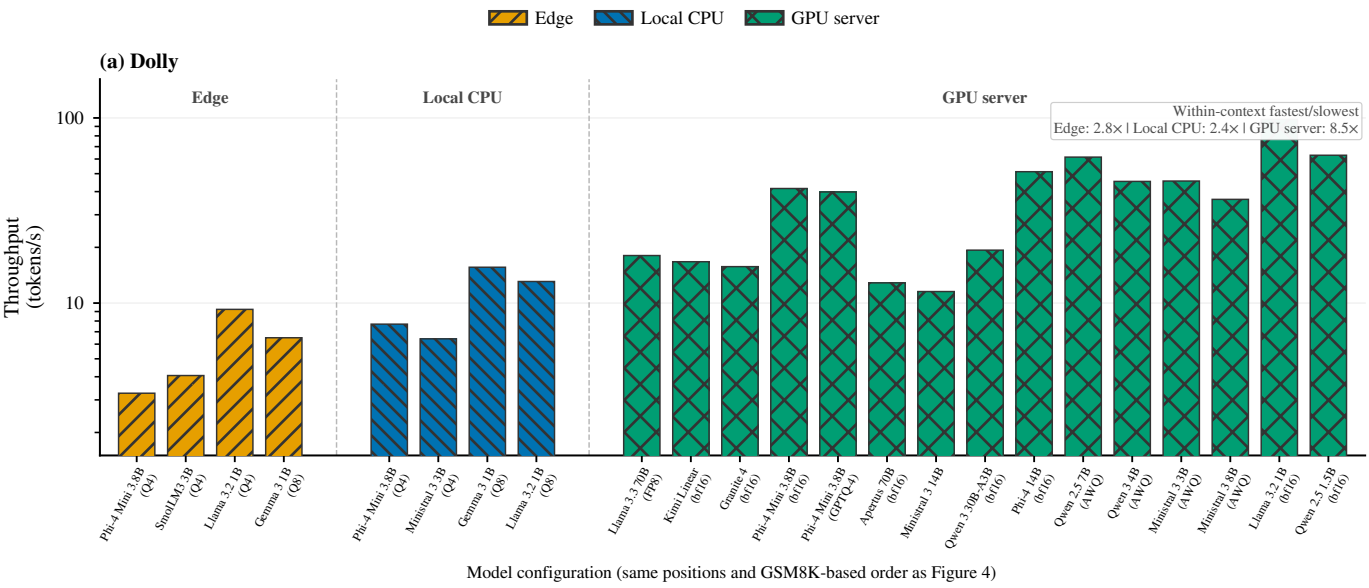
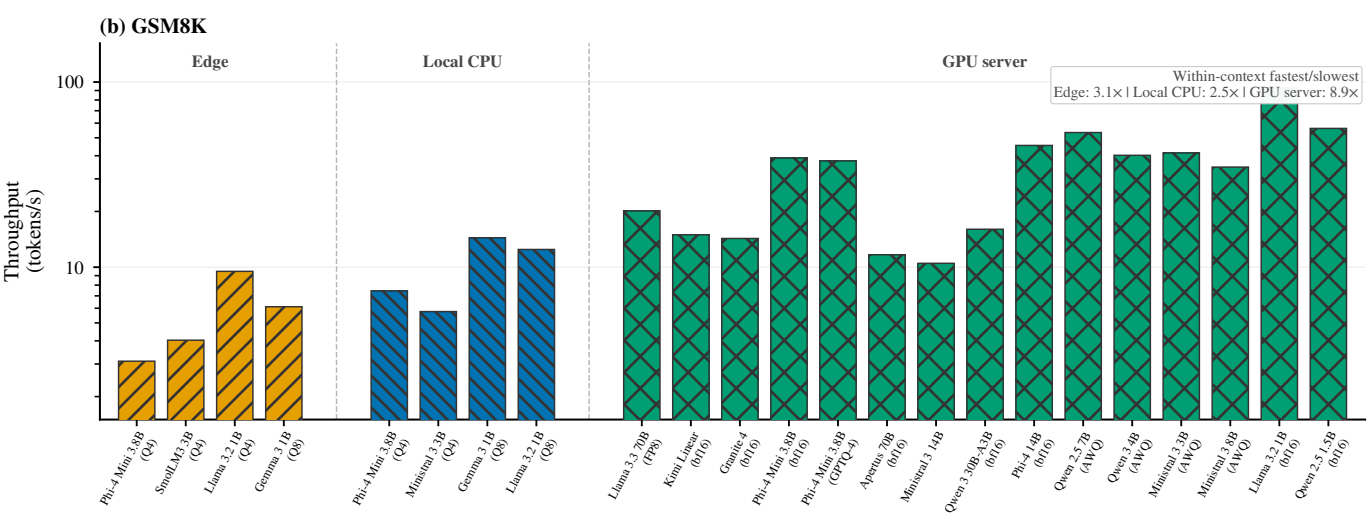
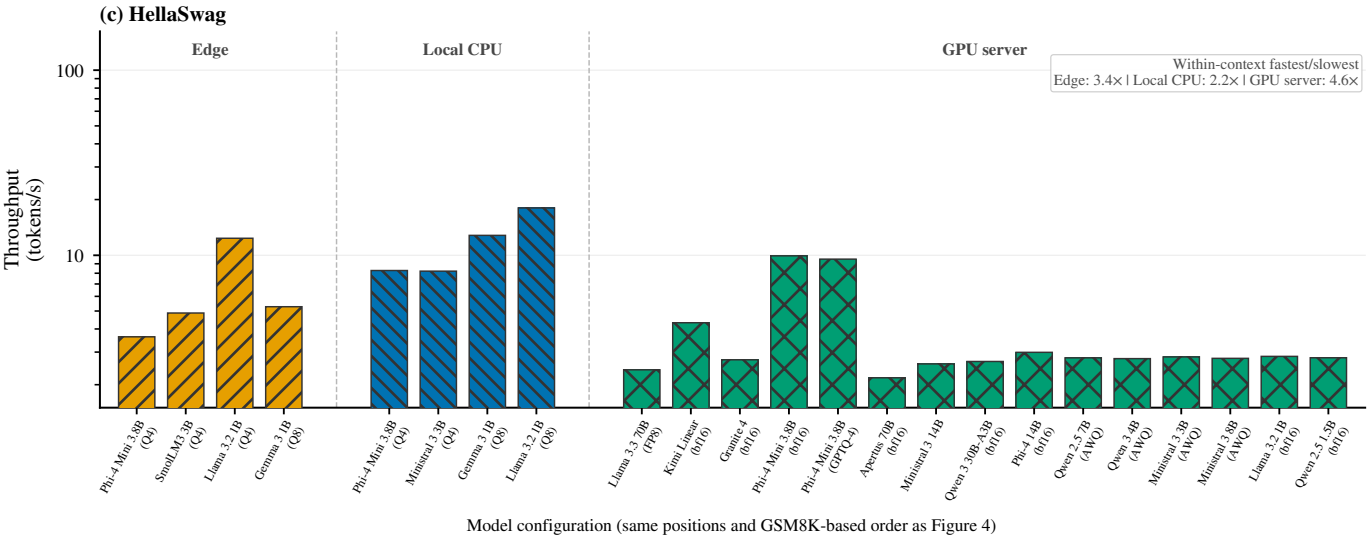
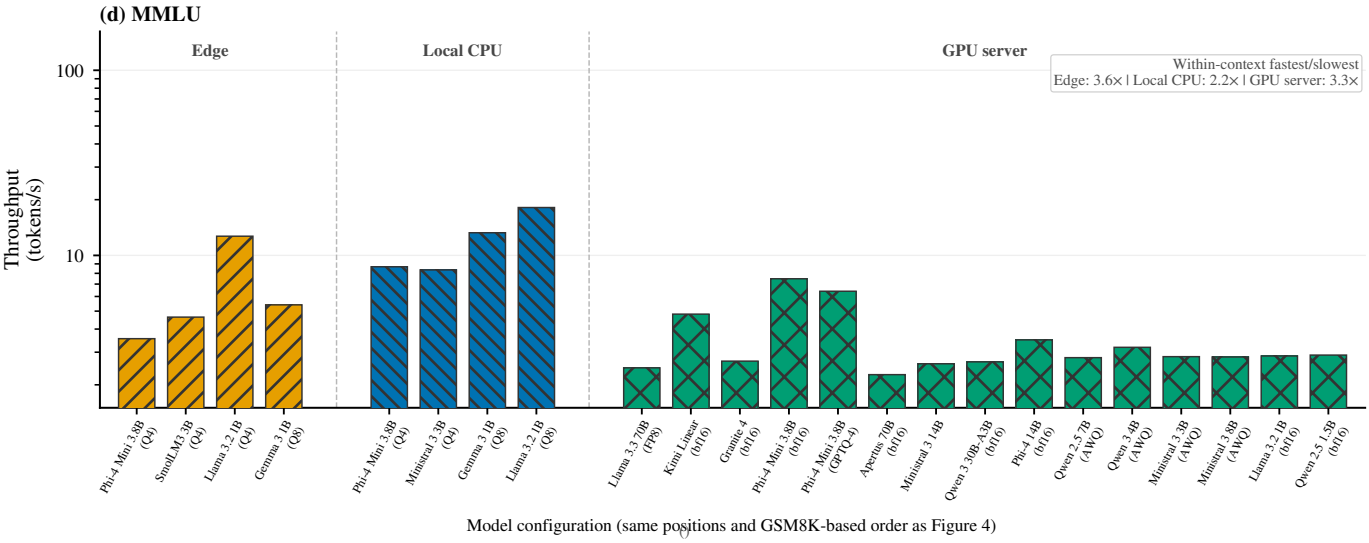

Edge
Local CPU
GPU server
(a) Dolly
Edge
Local CPU
GPU server
Within-context fastest/slowest
Edge: 2.8× | Local CPU: 2.4× | GPU server: 8.5×
Throughput (tokens/s)
100
10
Phi-4 Mini 3.8B (Q4)
SmolLM3 3B (Q4)
Llama 3.2 1B (Q4)
Gemma 3 1B (Q8)
Phi-4 Mini 3.8B (Q4)
Ministral 3 3B (Q4)
Gemma 3 1B (Q8)
Llama 3.2 1B (Q8)
Llama 3.3 70B (FP8)
Kimi Linear (bf16)
Granite 4 (bf16)
Phi-4 Mini 3.8B (bf16)
Phi-4 Mini 3.8B (GPTQ-4)
Apertus 70B (bf16)
Ministral 3 14B
Qwen 3 30B-A3B (bf16)
Phi-4 14B (bf16)
Qwen 2.5 7B (AWQ)
Qwen 3 4B (AWQ)
Ministral 3 3B (AWQ)
Ministral 3 8B (AWQ)
Llama 3.2 1B (bf16)
Qwen 2.5 1.5B (bf16)
Model configuration (same positions and GSM8K-based order as Figure 4)


(b) GSM8K
Edge
Local CPU
GPU server
Within-context fastest/slowest
Edge: 3.1× | Local CPU: 2.5× | GPU server: 8.9×
Throughput (tokens/s)
100
10
Phi-4 Mini 3.8B (Q4)
SmolLM3 3B (Q4)
Llama 3.2 1B (Q4)
Gemma 3 1B (Q8)
Phi-4 Mini 3.8B (Q4)
Ministral 3 3B (Q4)
Gemma 3 1B (Q8)
Llama 3.2 1B (Q8)
Llama 3.3 70B (FP8)
Kimi Linear (bf16)
Granite 4 (bf16)
Phi-4 Mini 3.8B (bf16)
Phi-4 Mini 3.8B (GPTQ-4)
Apertus 70B (bf16)
Ministral 3 14B
Qwen 3 30B-A3B (bf16)
Phi-4 14B (bf16)
Qwen 2.5 7B (AWQ)
Qwen 3 4B (AWQ)
Ministral 3 3B (AWQ)
Ministral 3 8B (AWQ)
Llama 3.2 1B (bf16)
Qwen 2.5 1.5B (bf16)
Model configuration (same positions and GSM8K-based order as Figure 4)


(c) HellaSwag
Edge
Local CPU
GPU server
Within-context fastest/slowest
Edge: 3.4× | Local CPU: 2.2× | GPU server: 4.6×
Throughput (tokens/s)
100
10
Phi-4 Mini 3.8B (Q4)
SmolLM3 3B (Q4)
Llama 3.2 1B (Q4)
Gemma 3 1B (Q8)
Phi-4 Mini 3.8B (Q4)
Ministral 3 3B (Q4)
Gemma 3 1B (Q8)
Llama 3.2 1B (Q8)
Llama 3.3 70B (FP8)
Kimi Linear (bf16)
Granite 4 (bf16)
Phi-4 Mini 3.8B (bf16)
Phi-4 Mini 3.8B (GPTQ-4)
Apertus 70B (bf16)
Ministral 3 14B
Qwen 3 30B-A3B (bf16)
Phi-4 14B (bf16)
Qwen 2.5 7B (AWQ)
Qwen 3 4B (AWQ)
Ministral 3 3B (AWQ)
Ministral 3 8B (AWQ)
Llama 3.2 1B (bf16)
Qwen 2.5 1.5B (bf16)
Model configuration (same positions and GSM8K-based order as Figure 4)


(d) MMLU
Edge
Local CPU
GPU server
Within-context fastest/slowest
Edge: 3.6× | Local CPU: 2.2× | GPU server: 3.3×
Throughput (tokens/s)
100
10
Phi-4 Mini 3.8B (Q4)
SmolLM3 3B (Q4)
Llama 3.2 1B (Q4)
Gemma 3 1B (Q8)
Phi-4 Mini 3.8B (Q4)
Ministral 3 3B (Q4)
Gemma 3 1B (Q8)
Llama 3.2 1B (Q8)
Llama 3.3 70B (FP8)
Kimi Linear (bf16)
Granite 4 (bf16)
Phi-4 Mini 3.8B (bf16)
Phi-4 Mini 3.8B (GPTQ-4)
Apertus 70B (bf16)
Ministral 3 14B
Qwen 3 30B-A3B (bf16)
Phi-4 14B (bf16)
Qwen 2.5 7B (AWQ)
Qwen 3 4B (AWQ)
Ministral 3 3B (AWQ)
Ministral 3 8B (AWQ)
Llama 3.2 1B (bf16)
Qwen 2.5 1.5B (bf16)
Model configuration (same positions and GSM8K-based order as Figure 4)